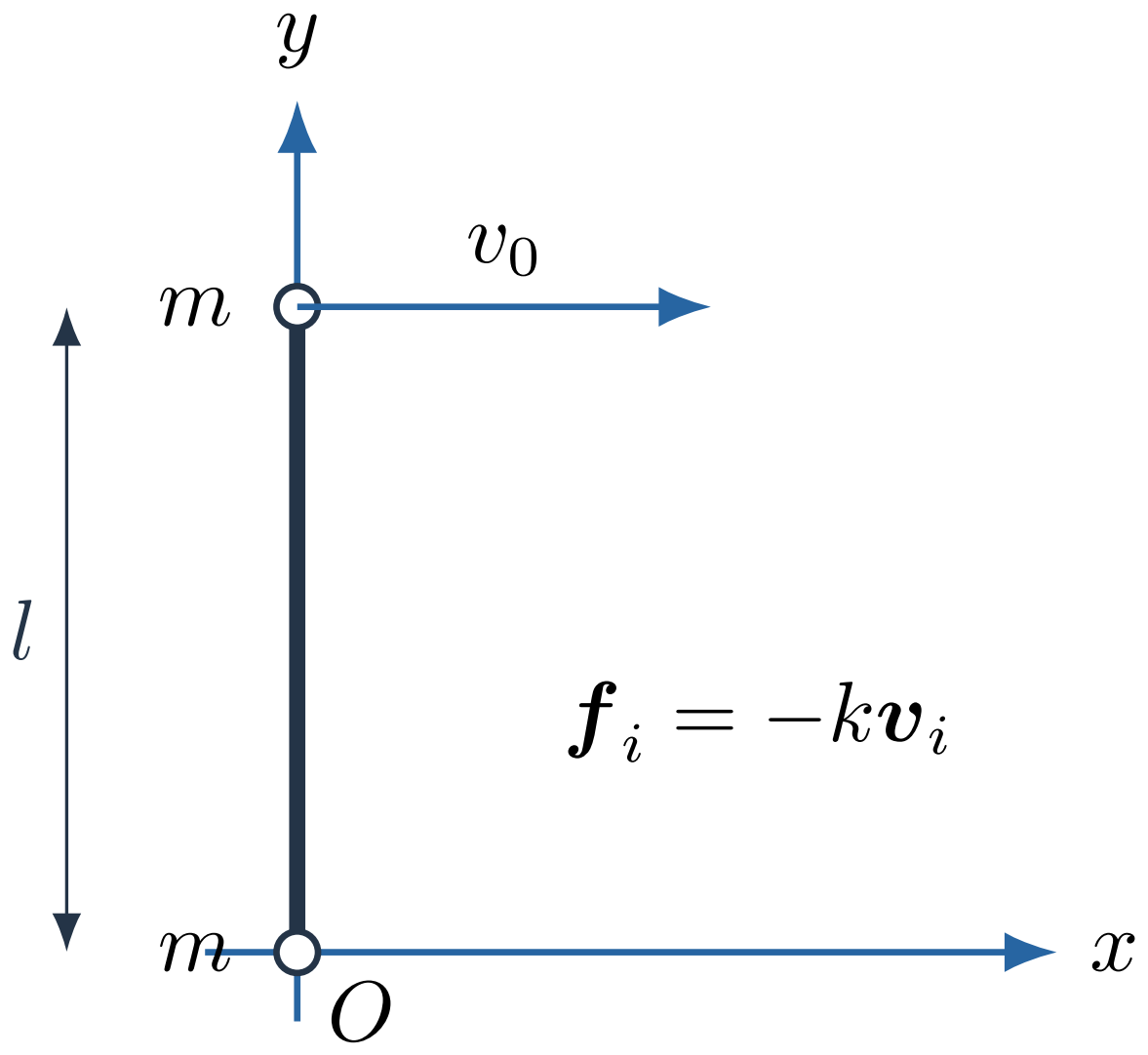
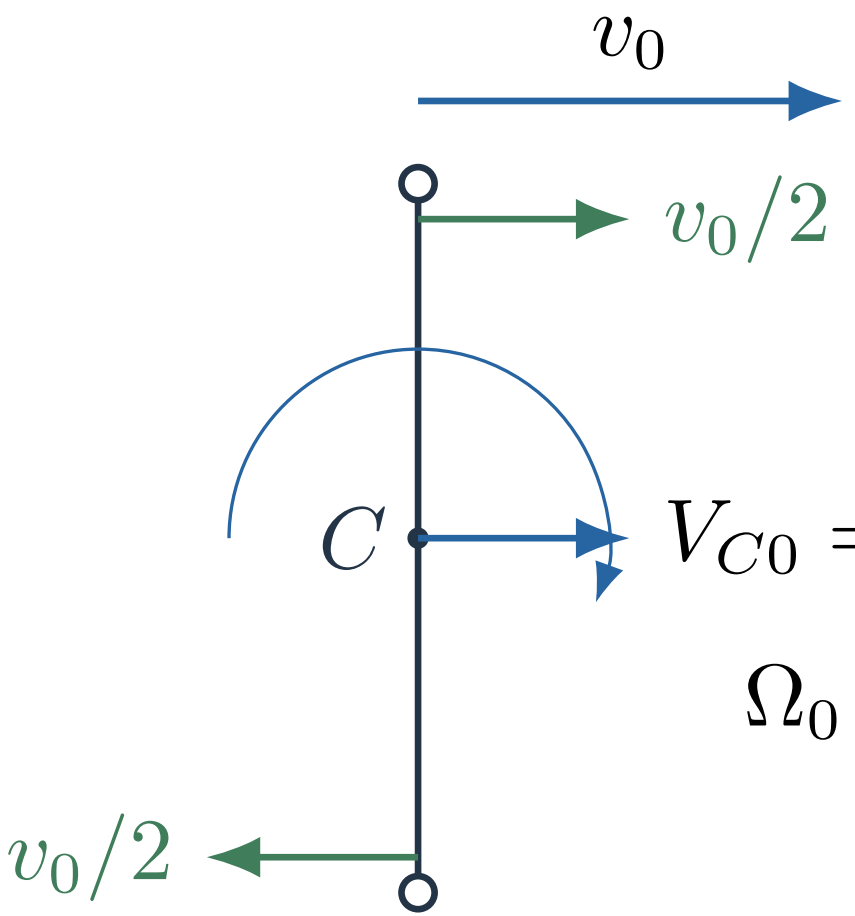

y
v_0
m
l
$\boldsymbol{f}_i = -k\boldsymbol{v}_i$
m
x
O

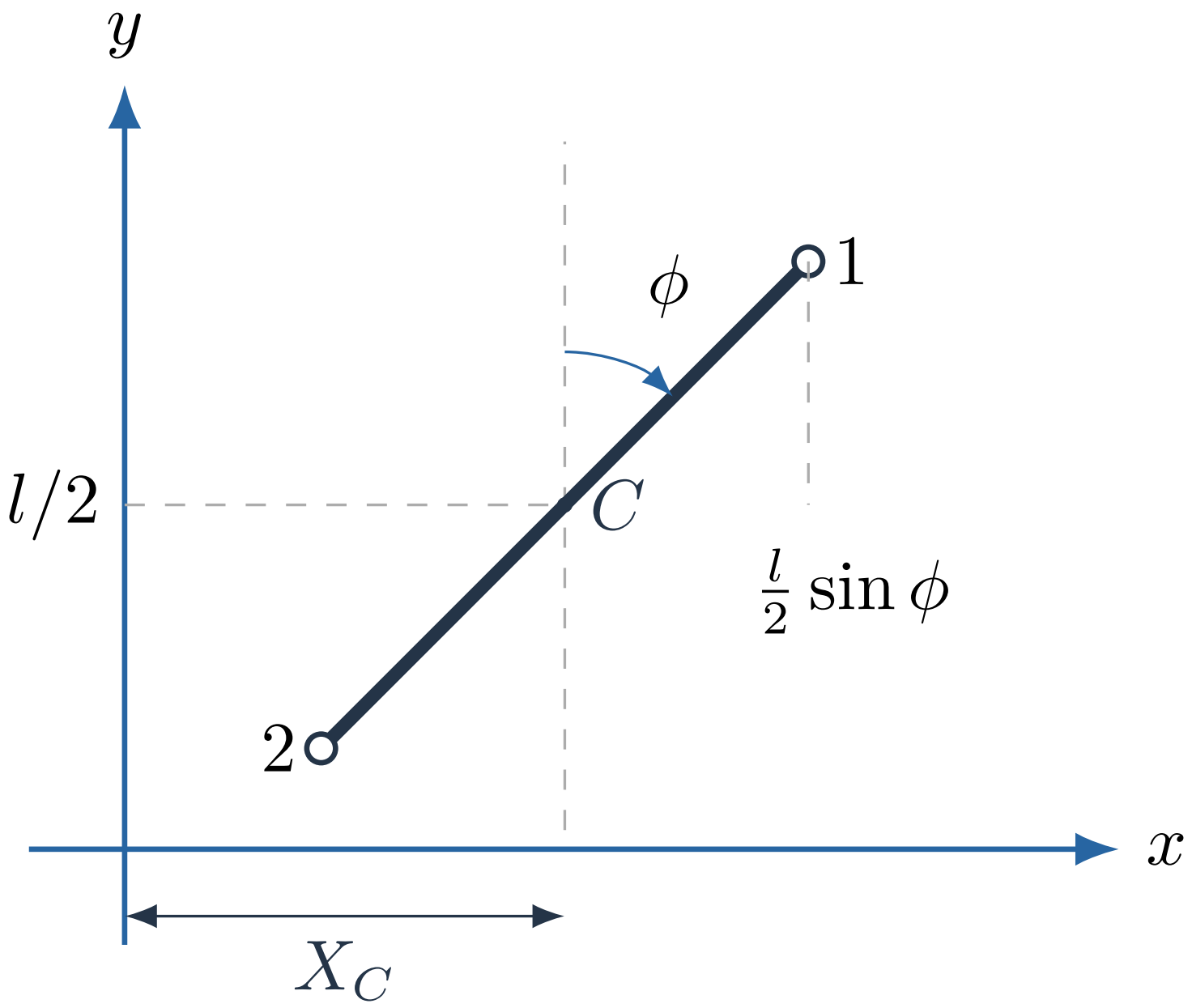


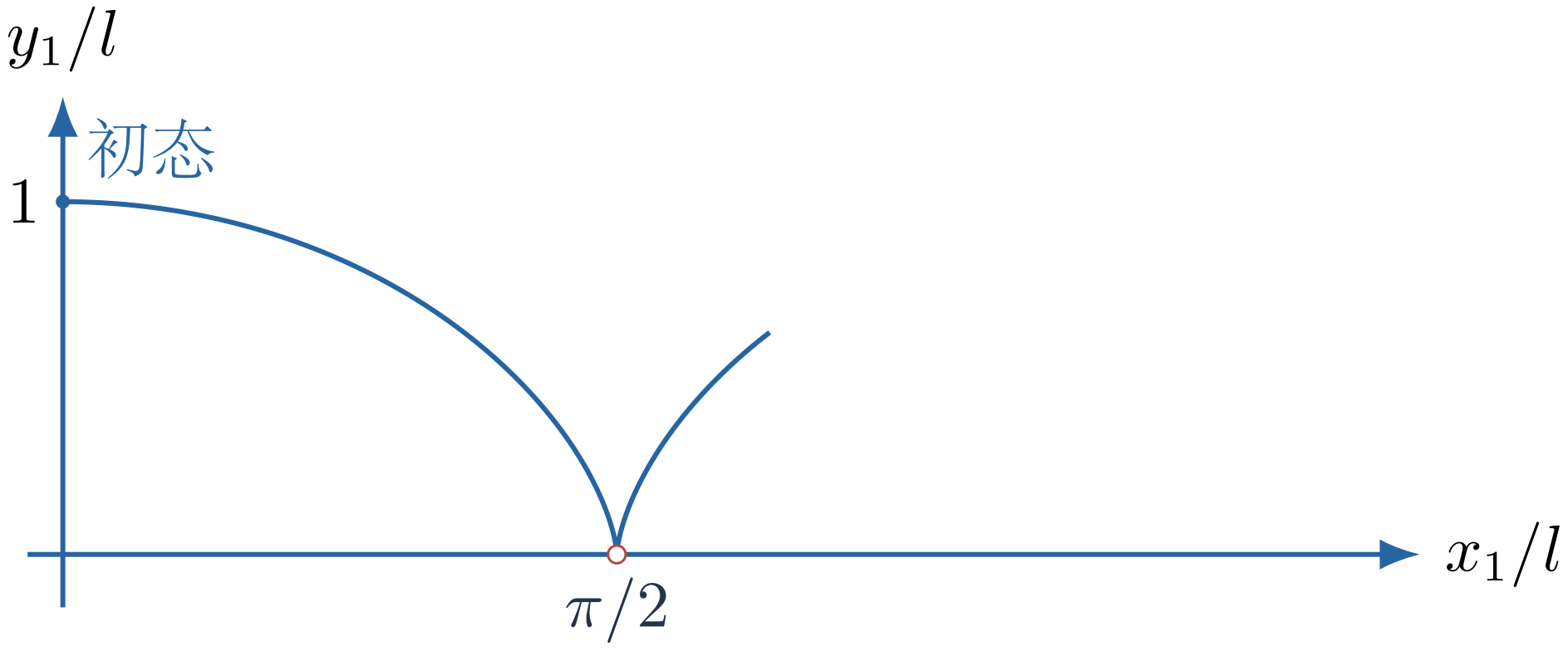
上端相对转动

$V_{C0} = v_0/2$

$\Omega_0 = -v_0/l$

下端合速度为零


y
x
1
2
C
ϕ
$l/2$
$\frac{l}{2}\sin\phi$
X_C



$\phi = \pi$： 瞬时零速